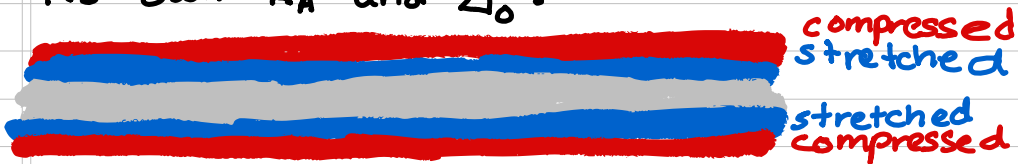


"Pressure Profile"

Complication

Membranes are not homogeneous, unlike the bending object from "bending vs stretching".

We can still imagine them as many thin sheets stacked on top of each other - but each one has its own k_A and Σ_0 :



stretching free energy at height z = $f_s(z) = \frac{k_A(z) (\Sigma - \Sigma_0(z))^2}{2 \Sigma_0(z)}$

surface tension at height z = $\gamma(z) = \frac{k_A(z) (\Sigma - \Sigma_0(z))}{\Sigma_0(z)}$

To get the total, integrate over z :

$$\begin{aligned} F_s &= \int \frac{k_A(z) (\Sigma - \Sigma_0(z))^2}{2 \Sigma_0(z)} dz = \int \frac{\gamma(z)}{2} [\Sigma - \Sigma_0(z)] dz \\ &= \underbrace{\frac{\Sigma}{2} \int \gamma(z) dz}_{\gamma, \text{ the overall surface tension}} - \int \frac{\gamma(z) \Sigma_0(z)}{2} dz \end{aligned}$$

Note that if the equilibrium area Σ_0 is independent of z :

$$F_s = \frac{\gamma}{2} \Sigma - \frac{\gamma}{2} \Sigma_0 \leftarrow \text{again, } \gamma = \int dz \gamma(z) \\ = \frac{\gamma}{2} (\Sigma - \Sigma_0)$$

so all the inhomogeneities get integrated to provide an effective surface tension and a simple macroscopic expression.

For this reason, it's common to calculate

$\gamma(z)$, the "pressure profile"

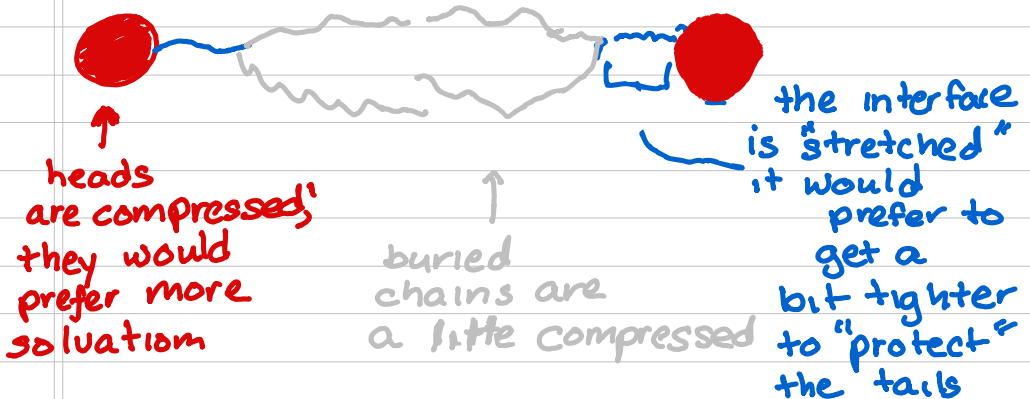
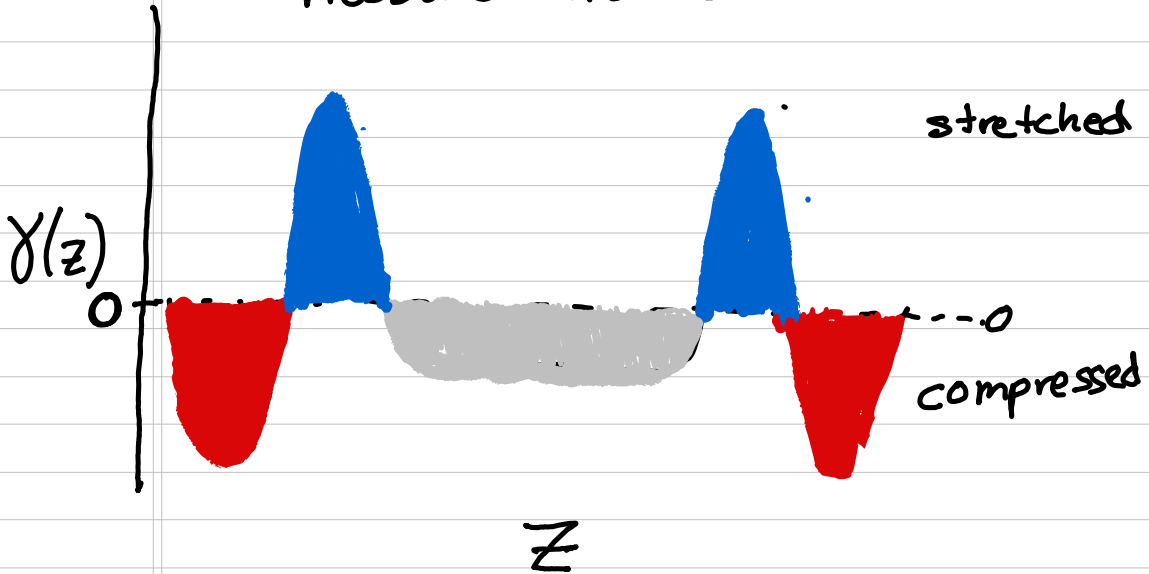
a horrible name; this is really a "surface tension profile," not "pressure"... but -

$\gamma(z)$ - or γ overall - ~~is~~ usually calculated using the pressure tensor:

$$P = \begin{pmatrix} P_{xx} & P_{xy} & P_{xz} \\ P_{yx} & P_{yy} & P_{yz} \\ P_{zx} & P_{zy} & P_{zz} \end{pmatrix} \text{ and } \gamma = P_{zz} - \frac{1}{2}(P_{xx} + P_{yy})$$

for symmetric lipid bilayers -
 $\gamma(z)$ looks something like this:

"Pressure Profile"



Now, for a symmetric bilayer without external tension: $\int \gamma(z) dz = 0$

but obviously that doesn't require that $\gamma(z) = 0$

everywhere. $\gamma(z)$ is peaked at the interface between oil and water, aka the "interfacial tension".

Bending from stretching II

So, last time we showed that for a homogeneous sheet,

$$F_{\text{bend}} = \frac{2 K_A \Sigma_0 H^2}{t} \int_{-t/2}^{t/2} z^2 dz$$

but for our real lipid bilayer,

$$\Sigma_0 \rightarrow \Sigma_0(z) \text{ and } \frac{K_A}{t} \rightarrow k_A(z)$$

$$F_{\text{bend}} = 2H^2 \int_{-t/2}^{t/2} z^2 k_A(z) \Sigma_0(z) dz$$

$$F_{\text{bend}} = 2H^2 \int_{-t/2}^{t/2} z^2 k_A(z) \Sigma'_0(z) dz$$

Special case 1: $k_A(z) = k_A/t$

$$F_{\text{bend}} = 2H^2 \frac{k_A}{t} \int_{-t/2}^{t/2} z^2 \Sigma_0(z) dz = 2k_c H^2 \Sigma$$

Σ_0 depends on z but k_A doesn't

$$k_c = \frac{k_A}{t} \int_{-t/2}^{t/2} z^2 \frac{\Sigma_0(z)}{\Sigma} dz \left. \vphantom{\int_{-t/2}^{t/2}} \right\} \begin{array}{l} k_c \text{ depends} \\ \text{on the} \\ \text{distribution} \\ \text{of equilibrium} \\ \text{areas in the} \\ \text{bilayer.} \end{array}$$

Special case 2: $\Sigma_0(z) = \Sigma_0 = \Sigma$

$$F_{\text{bend}} = 2H^2 \Sigma_0 \int_{-t/2}^{t/2} z^2 k_A(z) dz = 2k_c H^2 \Sigma_0$$

k_A depends on z but Σ_0 doesn't

$$k_c = \int_{-t/2}^{t/2} z^2 k_A(z) dz \left. \vphantom{\int_{-t/2}^{t/2}} \right\} \begin{array}{l} k_c \text{ depends} \\ \text{upon the} \\ \text{distribution} \\ \text{of stretching} \\ \text{energies in} \\ \text{the bilayer} \end{array}$$